Pearson's correlation coefficient

1

a Strong positive linear relationship

b Weak positive linear relationship

c No linear relationship

d Weak negative linear relationship

e Strong negative linear relationship

2

a Between 0.8 and 1

b Between 0.4 and 0.6

c Between -0.2 and 0.2

d Between -0.6 and -0.4

e Between -1 and -0.8

3

a Strong positive linear relationship

b Moderate positive linear relationship

c Weak positive linear relationship

d No linear relationship

e Weak negative linear relationship

f Moderate negative linear relationship

4

a As the heights of women increases, the probability of being turned down for a promotion decreases.

b As the population of a city increases, the number of speeding fines recorded increases.

c There is no evidence of a linear relationship between length of hair and length of fingernails.

d As the number of bylaws a council has about dog breeding increases, the number of dogs available for adoption at the local shelter tends to increase.

5 The second pair of data sets have a stronger correlation.

6

a A perfect positive correlation

b No relationship

c A perfect negative correlation

7

a 1

b A decimal between $-0.9 \text{ and } -1$ is acceptable.

c A decimal around 0 is acceptable.

d A decimal between $-0.3 \text{ and } -0.5$ is acceptable.

8

a

i Yes

ii Between x and z.

b

i Yes

ii Between x and y.

c

i No

ii Between x and y.

d

i No

ii Between x and t.

9 There is a strong, negative relationship between the life expectancy of a group of men and cigarettes they smoke a day.

10 There is a weak, negative linear relationship between the number of years in education a person completes and the number of pets.

11

a Non-linear

b A positive decimal around $r = 0.05$ is acceptable.

12

a Non-linear

b A negative decimal close to $r = -0.02$ is acceptable.

13

a Quadratic

b A negative decimal around $r = -0.68$ is acceptable.

Correlation vs causation

14

a Yes **b** No **c** No **d** No

e No **f** Yes **g** No **h** No

15

a No **b** Yes

16

a No **b** Yes **c** No **d** No

17

- **a** No
- **b** No
- **c** Yes
- **d** No

18

- **a** Greater than zero
- **b** Yes, as temperature rises this would cause people to buy more fans.

19

- **a** No. The high temperature may increase the number of people that go to the beach. This then increases the number of people that could drown.
- **b** No. There is an indirect relationship between the variables.

20 No

21

- **a** No
- **b** Yes, there are no other contributing factors or reasonable arguments to be made for the strong positive association between the approximate number of pirates out at sea and the average world temperature.
- **c** No

22

- **a** Set A
- **b** Set A, the points in the scatter plot appear to follow a curved path.
- **c** It appears that y can change significantly without much corresponding change in x.
- **d**
  - **i** Yes
  - **ii** Yes
  - **iii** Yes
  - **iv** No